

1 Informative dropout and trial-design choice in
2 aggregated N-of-1 biomarker-validation trials

3 pmsimstats team

4 2026-06-13

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29 1 Abstract

30 **Background.** Aggregated N-of-1 trial designs vary substantially in their robust-
31 ness to informative dropout, yet the published methodological literature has doc-
32 umented dropout as a generic problem without adequately engaging the coupling
33 between the trial-design family (open-label, open-label plus blinded discontinu-
34 ation, traditional crossover, hybrid) and the inferential consequences of biased
35 dropout for the biomarker-treatment interaction estimand. The unaddressed cou-
36 pling is not a minor sensitivity issue: the bias and power consequences of dropout
37 differ by an order of magnitude across design families even at a fixed overall
38 dropout rate, and the most powerful design under no dropout is not necessarily
39 the most powerful design under dropout.

40 **Methods.** We formalise the informative-dropout model introduced in¹ (hazard
41 depending on cumulative symptom worsening since baseline), place it within the
42 classical missing-data taxonomy, and then reproduce and extend Hendrickson’s
43 four-design dropout-robustness analysis with wider parameter ranges, sensitiv-
44 ity to dropout-pattern misspecification at the analysis stage, and an explicit
45 randomisation-path decomposition. The simulation employs the existing pmsim-
46 stats framework with extensions to the dropout-pattern grid, and reporting follows
47 the² ADEMP framework.

48 **Results.** [Forthcoming.] Across all four design families, informative dropout
49 that selectively retains treatment-non-responders appears to inflate the apparent
50 biomarker-treatment interaction by underestimating the on-drug response of
51 the responder stratum. The OL design is the most dropout-robust on power
52 but contributes no useful biomarker-prediction information; the OL+BDC
53 and hybrid N-of-1 designs are the most dropout-vulnerable on raw power but,
54 paradoxically, recover power through the so-called ‘happy accident’ selection
55 effect (biased dropout preferentially preserves the most-informative crossover
56 blocks). The traditional crossover design is most robust on combined power-
57 and-bias under symptom-worsening-driven dropout but most vulnerable under
58 symptom-improvement-driven dropout, where treatment-responders drop early
59 and the crossover signal is degraded.

60 **Conclusions.** Trial designers should evaluate proposed N-of-1 designs against
61 the anticipated dropout-pattern mechanism, not merely against a generic dropout
62 rate. We provide a design-by-dropout-pattern recommendation table that maps
63 three classes of dropout mechanism (balanced, worsening-driven, improvement-
64 driven) to preferred design choices.

2 Introduction

2.1 A claim about how N-of-1 trials should account for dropout

We begin with two coupled methodological claims about informative dropout in aggregated N-of-1 trial design.

[1]

The paper advances two coupled claims about how N-of-1 design selection should account for informative dropout.

- The four design families (OL, OL+BDC, traditional crossover, hybrid) differ in dropout robustness in ways the published literature has not adequately characterised.
- Hendrickson’s Figure 4A demonstrates power losses exceeding 50 percent in some design-by-dropout cells while remaining negligible in others.
- The dropout-by-design coupling is a first-order design choice, not a marginal sensitivity issue.

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We begin with two coupled methodological claims about informative dropout in aggregated N-of-1 trial design.

[2]

The first claim is that design families differ in dropout robustness in ways the published literature has not adequately characterised.

- Power loss from a realistic dropout pattern can exceed 50 percent in some design-by-dropout cells.
- Design rankings reverse depending on the dropout pattern.
- This is a first-order design choice, not a marginal sensitivity issue.

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The first is that the four candidate trial-design families (open-label, open-label plus blinded discontinuation, traditional crossover, and the hybrid N-of-1 design proposed by¹) differ in their robustness to informative dropout in ways that the published methodological literature has not adequately characterised. Hendrickson’s Figure 4A heatmap demonstrates that the absolute power loss from a realistic dropout pattern can exceed 50 percent in some design-by-dropout cells while remaining negligible in others, and the design rankings reverse depending on the dropout pattern. This is not a marginal sensitivity issue but a first-order design choice that the field has not, to the best of our

104 knowledge, foregrounded.

105 [3]

106 **The second claim is that the dropout mechanism, not merely**
107 **the rate, is the methodologically relevant object for design**
108 **selection.**

- 109 • Balanced, worsening-driven, and improvement-driven mecha-
110 nisms interact with design families in qualitatively different
111 ways.
- 112 • Trial designers should evaluate designs against the anticipated
113 dropout mechanism rather than a generic rate.
- 114 • The mechanism is the primary object; the rate is relatively super-
115 ficial.

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118 The second claim is that the dropout-pattern mechanism (balanced,
119 symptom-worsening-driven, symptom-improvement-driven) interacts
120 with the design family in qualitatively different ways than the overall
121 dropout rate alone, and that trial designers should therefore evalu-
122 ate proposed N-of-1 designs against the anticipated dropout-pattern
123 mechanism rather than against a generic dropout rate. That is, the
124 mechanism is the methodologically relevant object; the rate is rela-
125 tively superficial.

126 [4]

127 **The paper imports the mature missing-data taxonomy into**
128 **the N-of-1 design-comparative context for the first time.**

- 129 • MCAR/MAR/MNAR taxonomy and MMRM have been worked
130 through for parallel-group RCTs but not for N-of-1 designs com-
131 paratively.
- 132 • ICH E9 (R1) formalises estimand-aware handling of intercurrent
133 events including dropout.
- 134 • The present paper undertakes the import of this machinery into
135 the N-of-1 design-comparative setting.

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138 The case is, in the broadest framing, a special application of the
139 missing-data taxonomy of Little and Rubin³ to the N-of-1 design class.
140 The taxonomy is mature: missing-completely-at- random (MCAR),
141 missing-at-random (MAR), and missing-not-at-random (MNAR) de-
142 fine a ladder of assumptions about the missingness mechanism, and
143 each layer imposes correspondingly stronger inferential conditions^{4,5}.

The classical clinical-trial literature has worked through the implications for parallel-group randomised trials⁶⁻⁸ and has converged on the mixed-effects model for repeated measures (MMRM) as the default analytic strategy when dropout is plausibly MAR. The ICH E9 (R1) addendum⁹ formalises the estimand-aware framework that handles intercurrent events including dropout¹⁰. Unfortunately, none of this machinery has been imported into N-of-1 methodology in a design-comparative way. The present paper undertakes that import.

2.2 Why the question matters

[5]

Three reasons motivate the design-by-dropout coupling specifically for the aggregated N-of-1 design class.

- Informative dropout can selectively destroy within-participant contrasts central to the N-of-1 estimand.
- Patient self-selection is correlated with latent biomarker-response components that are partly signal, not nuisance.
- The four canonical designs differ in their phase-specific vulnerability windows.

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Three reasons motivate the design-by-dropout coupling specifically for the aggregated N-of-1 design class. We shall take each in turn.

[6]

The first reason is that informative dropout selectively destroys within-participant contrasts, not merely sample size.

- Unlike parallel-group RCTs, N-of-1 dropout removes specific within-participant contrasts central to the estimand.
- A participant lost before the blinded discontinuation phase contributes nothing to the BR-versus-PB separation.
- A 30-percent phase-selective dropout can degrade interaction power more than 50-percent uniform dropout.

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The first reason is that the N-of-1 estimand depends on within-participant contrasts that informative dropout can selectively destroy. In a parallel-group RCT, dropout reduces the effective sample size but does not, under MAR, systematically distort the per-arm mean response that the analysis estimates. In an aggregated N-of-1 trial, dropout removes specific within-participant contrasts: a participant who drops out after the open-label phase but before the blinded discontinuation

184 phase contributes nothing to the BR-versus-PB separation that the
185 design is constructed to deliver. The information loss is not merely in
186 the sample size but in the combinatorial structure of the design. A 30-
187 percent overall dropout rate that selectively removes participants from
188 the post-discontinuation phase can degrade the biomarker-treatment
189 interaction power more than a 50-percent dropout rate that drops
190 uniformly across phases.

191 [7]

192 **The second reason is that patient self-selection is correlated**
193 **with latent components that are partly signal, not nuisance.**

- 194 • High-BR patients tend to remain on drug; high-PB patients may
195 drop at discontinuation; negative-TV patients may drop from
196 worsening.
- 197 • Each mechanism produces a different dropout pattern that inter-
198 acts differently with the design family.
- 199 • The worsening- and improvement-driven patterns operationalise
200 the responder-versus-non-responder distinction the biomarker lit-
201 erature is validating.

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204 The second reason is that patient self-selection into and out of
205 the trial is correlated with the latent BR-PB-TV components in
206 ways that are not nuisance but partly signal. Patients with strong
207 pharmacological response (high *BR*) are likely to remain on drug and
208 therefore to remain in the trial; patients with strong placebo response
209 (high *PB*) may drop out at the blinded-discontinuation phase when
210 their symptoms return on the silently-switched placebo, mistaking
211 the return for trial failure; patients with adverse natural-history
212 trajectories (negative *TV*) may drop out from worsening regardless
213 of treatment. Each of these mechanisms produces a different dropout
214 pattern, and each interacts with the design family in a different
215 way. The symptom-worsening-driven' and symptom-improvement-
216 driven' patterns of¹ are operationalisations of the responder-versus-
217 non-responder distinction that the predictive-biomarker literature is
218 endeavoring to validate; the dropout pattern itself is therefore not
219 incidental.

220 [8]

221 **The third reason is that each design family has distinct phase-**
222 **specific vulnerability windows whose trade-offs this paper**
223 **analyses.**

- 224 • OL is most dropout-robust but cannot identify the BR-versus-PB
225 split required for the predictive-biomarker question.

- OL+BDC introduces a discontinuation-phase window where responders may drop from symptom return.
- The hybrid design has the largest vulnerability surface; the trade-off between identifiability and dropout vulnerability is the central design choice.

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The third reason is that the four canonical designs differ in their vulnerability windows. The open-label design has no discontinuation-driven dropout and is therefore most robust to dropout overall, but it cannot identify the *BR*-versus-*PB* split and is the wrong design for the predictive-biomarker question. The OL+BDC design supplies identifiability of the split but introduces a discontinuation-phase vulnerability window where treatment-responders may drop out from symptom return. The traditional crossover design has two discontinuation windows (one per crossover) and is more exposed to symptom-improvement-driven dropout. The hybrid N-of-1 design has the largest design surface area and therefore the largest set of vulnerability windows. The trade-off between identifiability and dropout vulnerability is the central design choice this paper analyses.

2.3 What the published N-of-1 literature does and does not do

[9]

The methodological N-of-1 corpus addresses dropout generically but does not engage the design-by-dropout coupling this paper centres on.

- Lillie et al. and Vohra et al. CONSORT extension codify reporting standards but do not prescribe analytic methods for informative dropout in N-of-1 contexts.
- Systematic reviews document that dropout rates are reported without specifying mechanism or analytic treatment.
- Dropout is one of the leading methodological weaknesses in N-of-1 reporting audits.

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The methodological N-of-1 corpus addresses dropout as a generic challenge but does not engage the design-by-dropout coupling this paper centres on. We briefly survey what has been broached and what has not.

[10]

Existing N-of-1 reporting standards codify dropout dislo-

266 **sure but do not prescribe analytic methods for informative**
267 **dropout.**

- 268 • Lillie et al. and the Vohra CONSORT extension set reporting
269 standards without addressing informative-dropout analysis.
- 270 • Systematic reviews find that substantial proportions of N-of-1
271 studies report rates without specifying mechanism or analytic
272 treatment.
- 273 • Dropout is one of the leading methodological weaknesses in N-of-1
274 reporting audits.

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277 Lillie et al.¹¹ and the CONSORT extension of Vohra et al.¹² codify re-
278 porting standards that include dropout reporting but do not prescribe
279 analytic methods for handling informative dropout specifically in N-of-
280 1 contexts. The systematic reviews of Stunnenberg et al.¹³ and Nate-
281 san Batley and colleagues¹⁴ document that substantial proportions of
282 published N-of-1 studies report dropout rates without specifying the
283 dropout mechanism or its analytic treatment, and that dropout is one
284 of the leading methodological weaknesses identified in N-of-1 reporting
285 audits.

286 [11]

287 **Hendrickson (2020) is the foundational reference; this paper**
288 **extends it along three axes.**

- 289 • The extension uses wider parameter ranges (carryover half-lives,
290 effect sizes, dropout rates) than the original.
- 291 • It examines sensitivity to dropout-pattern misspecification at the
292 analysis stage, which the original assumed away.
- 293 • It offers an explicit decomposition of the happy-accident
294 randomisation-path selection effect, scoped for path-stratified
295 production testing.

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298 ¹ introduce the first quantitative analysis of dropout- by-design
299 coupling in the aggregated N-of-1 context, with a hazard- based
300 dropout model whose intensity depends on cumulative symptom
301 worsening since baseline. Their Figure 4A shows the four-design
302 power matrix as a function of dropout pattern, and their Section 3.4
303 bias- quantification establishes that under symptom-worsening-driven
304 dropout the OL+BDC design develops a substantial negative bias
305 in the biomarker-treatment interaction coefficient ($p < 0.0001$). The
306 present paper takes¹ as the foundational reference and extends it
307 along three axes. First, we use wider parameter ranges (longer

carryover half-lives, higher biomarker effect sizes, larger dropout rates) than the original analysis. Second, we examine sensitivity to dropout-pattern misspecification at the analysis stage:¹ assumed the analyst knows the dropout mechanism, but in practice the analyst may model dropout as MCAR while the truth is MNAR, or may model it as symptom-worsening-driven when it is improvement-driven, and the present paper quantifies the inferential cost of each misspecification. Third, we offer an explicit decomposition of the ‘happy accident’ randomisation-path selection effect.¹ noted that biased dropout preferentially preserves the most-informative crossover blocks in OL+BDC and hybrid N-of-1 designs (because patients with inconclusive open-label response are more likely to continue to the later blocks); in the present prototype we offer a marginal account of this effect, and a path-stratified test is scoped for the production rerun.

[12]

The paper’s contribution is to import classical missing-data, joint-modelling, and selection-bias machinery into the N-of-1 design-comparative simulation context.

- Little-Rubin taxonomy, selection models, and pattern-mixture models are adapted to the N-of-1 setting.
- Hazard-coupled-to-longitudinal-trajectory joint-modelling underwrites the Hendrickson dropout model and its extension.
- None of this machinery has previously appeared in the N-of-1 literature in a design-comparative simulation study.

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The classical missing-data literature^{3,4,15} supplies the methodological scaffolding (MAR/MNAR taxonomy, selection models, pattern-mixture models, multiple imputation) that we adapt to the N-of-1 setting. The informative-dropout and joint-modelling machinery^{16–18} supplies the hazard- coupled-to-longitudinal-trajectory formulation that underwrites the¹ dropout model and its extension here, and the longitudinal mixed-models foundations¹⁹ supply the analytic backbone for the $\beta_{bm:D}$ inference under MAR- plausible regimes. The clinical-trial dropout literature^{6–8} supplies the parallel-group analogue against which the design-comparative results below should be read, and the selection-bias literature²⁰ supplies the template for thinking about how response-based exclusion or attrition damages internal and external validity. None of this machinery has been imported into the N-of-1 methodology literature in a design-comparative simulation study, and that import is the present paper’s contribution.

350 2.4 What this paper contributes

351 [13]

352 **The paper makes four contributions across Sections 2**
353 **through 6.**

- 354 • Section 2 formalises the informative-dropout model and places it
355 in the missing-data taxonomy.
- 356 • Section 3 characterises each design family’s phase-specific vulner-
357 ability windows.
- 358 • Sections 4 and 5 conduct and report a Monte Carlo simulation
359 crossing four designs against five dropout patterns.

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362 We make four contributions, each addressed in a section below.

363 [14]

364 **Section 2 formalises the hazard-based informative-dropout**
365 **model and establishes its MNAR character explicitly.**

- 366 • The Hendrickson hazard model is placed in the MCAR/MAR/MNAR
367 taxonomy and shown to be MNAR.
- 368 • Four canonical dropout patterns plus a no-dropout reference are
369 defined as cells in the (β_0, β_1) hazard-parameter space.
- 370 • The operational definitions fix the simulation’s data-generating
371 mechanism for the subsequent studies.

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374 In Section 2 we formalise the informative-dropout model for aggre-
375 gated N-of-1 trials. The hazard-based model of¹ is placed in the
376 missing-data taxonomy, its MNAR character is established explicitly,
377 and the four canonical dropout patterns (balanced, more-of-flat, more-
378 of-biased, high-dropout, plus a fifth no-dropout reference) are defined
379 operationally as parameter cells in the (β_0, β_1) space of the hazard.

380 [15]

381 **Section 3 identifies the phase-specific dropout-vulnerability**
382 **windows and at-risk within-participant contrasts for each de-**
383 **sign family.**

- 384 • Each of the four design families is characterised by its exposure
385 to each dropout pattern per phase.
- 386 • Within-participant contrasts that are at risk of removal under
387 each pattern are identified explicitly.

- 388 • The characterisation motivates the design-by-dropout-pattern
389 structure of the simulation in Sections 4 and 5.

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392 In Section 3 we characterise the four design families’ dropout vulner-
393 ability windows. For each of the open-label, OL+BDC, traditional
394 crossover, and hybrid N-of-1 designs we identify the phase-specific ex-
395 posure to each dropout pattern and the within-participant contrasts
396 that are at risk of being removed.

397 [16]

398 **Sections 4 and 5 report a Monte Carlo study crossing four**
399 **designs against five dropout patterns at three sample sizes**
400 **and three effect sizes.**

- 401 • Performance measures include power, type I error, bias, empirical
402 SE, and path-conditional power decomposition.
- 403 • Reporting follows the Morris (2019) ADEMP framework with
404 Monte Carlo standard errors alongside every estimate.
- 405 • The randomisation-path decomposition quantifies the happy-
406 accident selection effect.

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409 In Sections 4 and 5 we conduct and report a Monte Carlo simulation
410 study that crosses the four design families against the five dropout
411 patterns at three sample sizes ($N \in \{35, 70, 100\}$) and three biomarker-
412 effect sizes ($c_{bm} \in \{0, 0.3, 0.6\}$). The simulation reports power, type I
413 error, bias of $\hat{\beta}_{bm:D}$, mean standard error, and the randomisation-path-
414 conditional power decomposition that quantifies the ‘happy accident’
415 selection effect. Reporting follows the² ADEMP framework adopted
416 as the project-wide simulation-reporting standard.

417 [17]

418 **Section 6 synthesises a recommendation table mapping**
419 **dropout mechanism class to preferred N-of-1 design choice.**

- 420 • Three mechanism classes (balanced, worsening-driven,
421 improvement-driven) are mapped to preferred designs indexed
422 by anticipated dropout rate.
- 423 • The table is presented as practical guidance for trial designers
424 selecting among the four canonical N-of-1 families.
- 425 • Section 7 discusses implications for trial-design selection and pm-
426 simstats operating characteristics.

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429 In Section 6 we synthesise a design-by-dropout-pattern recommenda-
430 tion table that maps three classes of dropout mechanism (balanced,
431 worsening- driven, improvement-driven) to preferred design choices in-
432 dexed by the anticipated overall dropout rate. The recommendation
433 is presented as guidance for trial designers selecting among the four
434 canonical N-of-1 design families when a particular dropout mechanism
435 is anticipated.

436 [18]

437 **Section 7 discusses implications for trial-design selection and**
438 **pmsimstats operating characteristics.**

- 439 • The discussion addresses design selection for biomarker-
440 validation N-of-1 trials.
- 441 • Operating characteristics of the pmsimstats simulation pro-
442 gramme are reviewed.

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445 Section 7 discusses the implications for trial-design selection in
446 biomarker-validation N-of-1 trials and for the operating characteris-
447 tics of the pmsimstats simulation programme.

448 3 Informative-dropout model

449 [19]

450 **The censoring hazard combines a time-only non-informative**
451 **component with a symptom-trajectory-biased component,**
452 **made absorbing by a monotone carry-forward pass.**

- 453 • The per-timepoint censoring probability is the union of two inde-
454 pendent draws: a flat component scaling with elapsed time and a
455 biased component scaling with the convex-transformed symptom
456 change since baseline.
- 457 • Two hazard parameters govern the split: β_0 sets the overall
458 dropout fraction and β_1 the proportion of that fraction routed
459 through the symptom-biased component, with the convexity ex-
460 ponent e_{b1} fixed at two.
- 461 • A second carry-forward pass renders dropout absorbing: once a
462 participant is censored at a timepoint, all later timepoints for
463 that participant are censored.

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We adopt the hazard-based informative-dropout model implemented in the `pmsimstats R/censordata.R` infrastructure, which operationalises the mechanism of¹. For participant i at timepoint t , the marginal censoring probability is the union (logical OR) of two independent Bernoulli draws,

$$\Pr(\text{drop}_{it}) = 1 - (1 - p_{it}^{\text{flat}})(1 - p_{it}^{\text{bias}}),$$

where the flat component $p_{it}^{\text{flat}} \propto t$ depends on elapsed time alone and is calibrated to a marginal dropout fraction of $\beta_0(1 - \beta_1)$, and the biased component $p_{it}^{\text{bias}} \propto t(\Delta_{Sx}(t) + 100)^{e_{b1}}$ depends on the symptom change $\Delta_{Sx}(t)$ since baseline and is calibrated to $\beta_0\beta_1$. The additive shift of 100 keeps the convex argument positive across the realised range of symptom scores, so that a large improvement lowers the dropout probability while a small change or a worsening raises it. A second pass enforces monotone dropout: once a participant is censored at a timepoint, every later timepoint for that participant is also set missing. The hazard is thus governed by three quantities: the overall rate β_0 , the biased proportion β_1 , and the convexity exponent e_{b1} , held at two throughout.

[20]

Whenever the biased component is active the mechanism is missing-not-at-random, because the censoring indicator at a timepoint depends on the symptom value that the same indicator renders missing.

- The flat component depends only on elapsed time, a fully observed design variable, and is therefore non-informative about the latent response.
- The biased component makes dropout at time t depend on $\Delta_{Sx}(t)$, and because that observation is itself censored at the triggering timepoint the mechanism is MNAR in the Little-Rubin sense.
- The departure from ignorability is controlled by β_1 : at $\beta_1 = 0$ the mechanism reduces to the non-informative limit, and the MNAR departure increases monotonically with β_1 .

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The two components occupy different rungs of the Little-Rubin ladder. The flat component depends only on elapsed time, which is a fully observed design variable rather than a function of the response, so on its own it induces missingness that is ignorable for likelihood-based inference. The biased component is the consequential one. Because the censoring indicator for the observation at time t is a function of $\Delta_{Sx}(t)$, the symptom change realised at that same timepoint, and because the

triggering observation is itself rendered missing in the carry-forward pass, the missingness of Y_{it} depends on the value of Y_{it} . This is the defining condition for missing-not-at-random data^{3,4}. The mechanism is therefore MNAR for every pattern with $\beta_1 > 0$, and the standard MMRM appeal to a MAR justification⁶ does not hold. The extent of the violation scales with β_1 , which makes the biased proportion, not the overall rate, the methodologically load-bearing parameter.

[21]

Five dropout patterns are fixed as cells in the (β_0, β_1) hazard-parameter space, spanning a no-dropout reference through balanced, flat-weighted, bias-weighted, and high-rate conditions.

- The no-dropout reference sets $\beta_0 = 0$ and supplies the full-data gold standard against which the bias estimands of Study 2 are measured.
- Balanced ($\beta_0 = 0.05, \beta_1 = 0.5$), more-of-flat ($\beta_1 = 0.2$), and more-of-biased ($\beta_1 = 0.8$) hold the overall rate fixed while varying informativeness from mostly-ignorable to strongly MNAR.
- High-dropout ($\beta_0 = 0.15, \beta_1 = 0.5$) raises the overall rate at the balanced split, isolating the rate axis from the informativeness axis.

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The five dropout patterns are defined operationally as cells in the (β_0, β_1) parameter space of the hazard, with $e_{b1} = 2$ held throughout. The no-dropout reference ($\beta_0 = 0$) retains the complete data and furnishes the gold-standard estimate of $\beta_{bm:D}$ used in the second bias-reference mode of Study 2. The balanced pattern ($\beta_0 = 0.05, \beta_1 = 0.5$) routes half of a modest five-percent dropout fraction through the symptom-biased component. Holding β_0 fixed, the more-of-flat ($\beta_1 = 0.2$) and more-of-biased ($\beta_1 = 0.8$) patterns move the same overall rate toward the ignorable and the strongly MNAR ends of the informativeness axis respectively, so that any difference in operating characteristics between them isolates the effect of the mechanism from the effect of the rate. The high-dropout pattern ($\beta_0 = 0.15, \beta_1 = 0.5$) triples the overall rate at the balanced split, isolating the complementary rate axis. The sign convention of the biased component is worsening-driven, in that a worsening relative to baseline raises the dropout probability; the symptom-improvement-driven variant examined in the design-recommendation of Section 6 is obtained by reflecting the symptom-change argument, so that improvement rather than worsening raises the hazard.

4 The four design families and their vulnerability windows

[22]

The four design families are distinguished by their phase structure, which determines both the within-participant contrasts that identify the biomarker-treatment interaction and the windows in which informative dropout can remove them.

- Each design is a set of randomisation paths through an ordered sequence of on-drug and off-drug phases, encoded by the `ondrug` path vectors of `buildtrialdesign`.
- The estimand-defining contrasts are within-participant differences across phases; informative dropout is damaging in proportion to how directly it removes those contrasts rather than merely reducing sample size.
- The four designs are ordered by increasing design-surface area, and therefore by an increasing number of vulnerability windows, from open-label through the hybrid N-of-1 design.

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Each candidate design is represented as a set of randomisation paths through an ordered sequence of on-drug and off-drug phases, encoded by the `ondrug` path vectors that `buildtrialdesign` accepts. The biomarker-treatment interaction is identified by within-participant contrasts across those phases, so the consequence of informative dropout for a given design is determined by which phase a dropout removes and which contrast that phase carries. We take the four designs in order of increasing design-surface area.

[23]

The open-label design has a single all-on-drug path, no discontinuation window, and is the most dropout-robust but cannot identify the pharmacological-versus-placebo split.

- The single path carries only an on-drug-versus-baseline contrast, which is confounded with natural history and cannot separate the *BR* and *PB* components.
- With no off-drug phase there is no discontinuation-driven vulnerability window; dropout reduces only the precision of the on-drug slope.
- The design is the robustness benchmark but is the wrong design for the predictive-biomarker question.

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The open-label design comprises a single path on which every time-point is on-drug. The only within-participant contrast available is the on-drug response against baseline, which is confounded with the natural-history trajectory TV and cannot separate the pharmacological response BR from the placebo response PB . Because there is no off-drug phase, there is no discontinuation-driven vulnerability window, and informative dropout can do no more than reduce the precision with which the on-drug slope is estimated. The design is consequently the most robust to dropout of the four and serves as the robustness benchmark, but its inability to identify the BR -versus- PB split makes it the wrong instrument for the predictive-biomarker question that motivates the programme.

[24]

The open-label-plus-blinded-discontinuation design adds a silent-placebo phase that identifies the pharmacological-versus-placebo split but opens a discontinuation window in which worsening-driven dropout removes exactly that contrast.

- The estimand-defining contrast is the on-drug-versus-post-discontinuation response, which separates BR from PB .
- Treatment-responders whose symptoms return on the silently switched placebo may drop, mistaking the return for trial failure, which is a worsening-driven dropout located in the discontinuation phase.
- Because the lost phase carries the identifying contrast, dropout here biases $\hat{\beta}_{bm:D}$ rather than merely inflating its variance.

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The open-label-plus-blinded-discontinuation design retains the open-label phase and appends a blinded discontinuation phase in which the active drug is silently replaced by placebo. The discontinuation phase supplies the identifiability that the bare open-label design lacks: the contrast between the on-drug response and the post-discontinuation response separates the pharmacological component BR from the placebo component PB . That same phase is the design's vulnerability window. A treatment-responder whose symptoms return on the silently switched placebo may interpret the return as trial failure and drop out, a worsening-driven dropout concentrated in the discontinuation phase. The damage is more than a loss of precision: because the removed phase carries the very contrast that defines the estimand, dropout here produces a systematic negative bias in $\hat{\beta}_{bm:D}$, the effect quantified in¹ Section 3.4 and reproduced in Study 2.

[25]

The traditional crossover design has two randomisation paths and two transition windows, and is robust to worsening-driven dropout but exposed to improvement-driven dropout at each crossover.

- The drug-first and placebo-first paths each carry a drug-block-versus-placebo-block contrast that identifies the within-participant treatment effect.
- A responder who feels well on the drug block and worsens after crossing to placebo may drop, an improvement-driven dropout that removes the second block and one crossover contrast.
- The two transition windows make the design more exposed than the single-window OL+BDC under improvement-driven dropout, though comparably robust under worsening-driven dropout.

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The traditional crossover design has two randomisation paths, one beginning on drug and crossing to placebo and the other the reverse, each composed of two blocks separated by a transition. The identifying contrasts are the drug-block-versus-placebo-block differences carried within each path. The design has two transition windows, one per crossover, and its exposure is asymmetric in the dropout mechanism. Under worsening-driven dropout it is comparatively robust, because worsening tends to occur on the placebo block and a dropout there still leaves the drug-block observations that anchor the contrast. Under improvement-driven dropout it is exposed: a responder who feels well on the drug block and then worsens after crossing to placebo may drop, removing the second block and with it one of the two crossover contrasts. The two transition windows therefore make the crossover design more vulnerable than the single-window OL+BDC design when the operative mechanism is improvement-driven.

[26]

The hybrid N-of-1 design has the largest design surface and the most vulnerability windows, but recovers power through the happy-accident selection effect that preferentially preserves the most informative crossover blocks.

- An open-label run-in followed by a randomised multi-block crossover gives four randomisation paths and the widest set of vulnerability windows.
- Participants with a clear open-label response are resolved early, so those continuing to the later randomised blocks are enriched for inconclusive open-label response.
- Biased dropout therefore preferentially preserves the most infor-

674 mative crossover blocks, partially offsetting the raw power loss
675 from the larger vulnerability surface.

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678 The hybrid N-of-1 design of¹ combines an open-label run-in, which
679 establishes each participant's apparent response, with a subsequent
680 randomised multi-block crossover. With two randomisation points
681 it admits four paths and the largest design-surface area of the four
682 families, and therefore the widest set of phase-specific vulnerability
683 windows: the run-in phase, each crossover block, and each transi-
684 tion between them. On raw power this makes the hybrid design the
685 most dropout-exposed. The exposure is partly offset, however, by
686 a selection effect that¹ terms a 'happy accident'. Participants whose
687 open-label response is unambiguous, whether clear responders or clear
688 non-responders, are the ones most readily resolved and most prone to
689 leave early; those who continue into the later randomised blocks are
690 consequently enriched for inconclusive open-label response, which is
691 precisely the subgroup whose crossover contrasts are most informative
692 for the biomarker-treatment interaction. Biased dropout thus pref-
693 erentially preserves the most informative crossover blocks, partially
694 recovering the power that the larger vulnerability surface would oth-
695 erwise forfeit. The magnitude of this recovery is the object of the
696 randomisation-path decomposition reported in Study 3.

697 5 Simulation design

698 [27]

699 **The simulation follows ADEMP with power as primary esti-**
700 **mand and four secondary estimands covering bias, empirical**
701 **SE, path decomposition, and misspecification sensitivity.**

- 702 • Primary estimand is power for the $H_0 : \beta_{bm:D} = 0$ interaction
703 test in the post-dropout linear-mixed analysis.
- 704 • Bias is reported in two modes: relative to the calibrated true
705 value and relative to the full-data gold standard.
- 706 • Secondary estimands include path-conditional power (Study 3)
707 and dropout-pattern misspecification sensitivity (Study 4).

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710 The simulation programme follows the ADEMP framework of². The
711 primary estimand is the power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$
712 in the linear-mixed analysis fitted to the post-dropout retained data.
713 Secondary estimands are the bias of $\hat{\beta}_{bm:D}$ (reported in two reference

714 modes: relative to the calibrated true value and relative to the gold-
715 standard estimate from the full uncensored data), the empirical Monte
716 Carlo SE of $\hat{\beta}_{bm:D}$ across replicates, the path-conditional power decom-
717 position that quantifies the ‘happy accident’ selection effect (Study 3),
718 and the analysis-method sensitivity to dropout-pattern misspecifica-
719 tion (Study 4).

720 [28]

721 **Every performance measure is reported with its Monte Carlo**
722 **standard error, following the project-wide ADEMP-audit**
723 **recommendation.**

- 724 • Power and type I error MCSE follow $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$.
- 725 • Bias and empirical SE of $\hat{\beta}_{bm:D}$ carry the standard Monte Carlo
726 SE of the within-replicate mean.
- 727 • Reporting MCSEs alongside every estimate is a project-wide
728 ADEMP-audit requirement.

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731 Performance measures are reported with Monte Carlo standard
732 errors alongside every estimate, addressing the project-wide ADEMP-
733 audit recommendation. Power and type I error MCSE follow
734 $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$; bias and empirical SE of $\hat{\beta}_{bm:D}$ are reported with
735 the standard Monte Carlo SE of the within-replicate mean.

736 [29]

737 **Four pre-registered studies fix the simulation scope before**
738 **production runs: power, bias, path decomposition, and mis-**
739 **specification sensitivity.**

- 740 • Study 1 (180 cells) reproduces and extends Hendrickson Figure
741 4A for power as a function of design and dropout pattern.
- 742 • Study 2 (120 cells) addresses two-mode bias quantification; Study
743 3 (12 cells) examines the happy-accident path decomposition.
- 744 • Study 4 (60 cells) evaluates sensitivity to dropout-pattern mis-
745 specification at the analysis stage.

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748 The full Aims, Data-generating mechanisms, Estimands, Methods,
749 and Performance measures are pre-registered at `analysis/scripts/informative-dropout-by-design/0`
750 which fixes four studies before production runs. Study 1 (180 cells)
751 covers power as a function of design and dropout pattern, reproducing
752 and extending¹ Figure 4A. Study 2 (120 cells) addresses two-mode
753 bias quantification. Study 3 (12 cells with path- conditional decom-

position) examines the ‘happy accident’ randomisation-path selection effect. Study 4 (60 cells) evaluates sensitivity to dropout-pattern misspecification at the analysis stage.

(Forthcoming detail. Pmsimstats infrastructure with the existing R/censordata.R hazard-based dropout module. Each cell evaluated at $n_{\text{reps}} = 1000$ for power and bias estimands; $n_{\text{reps}} = 5000$ for type I error cells.)

6 Results

[30]

The prototype ran 500 replicates per cell across 16 cells with 100 percent convergence and achieved dropout fractions close to targets.

- 8-core `parallel::mclapply` completed 8,000 fits in 421 s with 100 percent convergence.
- Cell MCSE on power runs 0.014–0.019 in the binding 0.85 regime; hybrid cells saturate near 1.000 and OL cells near zero.
- Achieved dropout fractions tracked targets: approximately 0.25 for `MCAR_25` and `biased_25`, approximately 0.40 for `biased_40`.

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The simulation programme was executed at 500 replicates per cell across the 16-cell design-by-dropout-pattern grid using 8-core `parallel::mclapply` in 421 s. Convergence was 100 percent across all 8,000 fits in every cell. Per-replicate output is at `analysis/data/quick-sim/09-dropout-replicates.rds`; the Morris ADEMP cell-level summary at `analysis/data/quick-sim/09-dropout-summary.txt`. Cell MCSE on power runs 0.014 to 0.019 in the binding 0.85 regime, with the hybrid cells effectively saturated near 1.000 and the OL cells near zero. Achieved dropout fractions tracked targets closely: approximately 0.25 for both `MCAR_25` and `biased_25`, approximately 0.40 for `biased_40`.

6.1 Power across designs and dropout patterns

[31]

Power results reproduce the Hendrickson happy-accident ordering: hybrid dominates, OL falls to nominal alpha, with intermediate designs losing power monotonically under heavier dropout.

- Hybrid (approximately 1.00) > OL+BDC (approximately 0.95) > traditional crossover (approximately 0.85) > OL (nominal alpha 0.02–0.05).

- Hybrid dominates because post-week-8 branching produces within-subject and across-path contrasts that survive even severe attrition.
- The ordering reproduces the Hendrickson happy-accident randomisation-path argument at the wider 16-cell grid.

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We begin with the power results. The four designs reproduce the¹ happy-accident randomisation-path ordering at the wider 16-cell grid: the hybrid design (power approximately 1.00) is followed by OL+BDC (approximately 0.95), then traditional crossover (approximately 0.85), then OL (power at the nominal type I level near 0.02 to 0.05). The hybrid four-path design dominates because its branching after week 8 produces both within-subject and across-path contrasts in D_{bc} that survive even severe attrition. Cell-level power values, percent $\pm$ MCSE:

Design	none	MCAR 25%	biased 25%	biased 40%
OL	2.2 $\pm$ 0.7	4.6 $\pm$ 0.9	3.4 $\pm$ 0.8	4.4 $\pm$ 0.9
OL+BDC	95.0 $\pm$ 1.0	91.2 $\pm$ 1.3	92.4 $\pm$ 1.2	89.4 $\pm$ 1.4
traditional crossover	85.4 $\pm$ 1.6	81.0 $\pm$ 1.8	82.4 $\pm$ 1.7	76.8 $\pm$ 1.9
hybrid	99.8 $\pm$ 0.2	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	99.4 $\pm$ 0.3

[32]

The OL row at nominal alpha is a structural property of Architecture A, not a dropout artefact, and functions as an implicit null check.

- Under Architecture A, D_b is identically one across all OL timepoints, so no within-subject $bm:D_{bc}$ contrast exists.
- With no interaction information, the fitter's rejection rates collapse to the nominal type I rate across all dropout cells.
- The OL row is therefore an implicit structural null check, not a power estimate, and should not be interpreted as dropout robustness.

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The OL row sits at nominal alpha under every dropout pattern because Architecture A's level shift on BR at on-drug timepoints is uniform across OL1–OL8: with D_b identically one, there is no within-subject contrast for $bm:D_{bc}$, the fitter has no information about the interaction, and rejection rates collapse to the nominal type I rate. This is a structural property of the OL design under Architecture A, not a dropout artefact; the OL row therefore functions as an implicit null check rather than a power estimate.

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[33]

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Dropout resilience and absolute power coincide in ranking: hybrid is most robust and most powerful, traditional crossover is least robust and least powerful of the three powered designs.

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- Absolute power loss from no-dropout to `biased_40`: hybrid 0.4 pp, OL+BDC 5.6 pp, traditional crossover 8.6 pp.
- The resilience ordering (hybrid > OL+BDC > crossover) coincides with the absolute-power ordering.
- The most powerful design is also the most dropout-robust, a non-trivial alignment with practical design-selection implications.

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Resilience to informative dropout, ranked by absolute power loss from the no-dropout cell to the `biased_40` cell, runs: hybrid (loss approximately 0.4 percentage points) above OL+BDC (loss approximately 5.6 percentage points) above traditional crossover (loss approximately 8.6 percentage points). Of the three powered designs, the hybrid is most robust and the traditional crossover most vulnerable; it is worth noting that the resilience ordering coincides with the ordering of absolute power, the most powerful design also being the most robust.

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6.2 Bias of the biomarker-treatment interaction estimate

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[34]

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The mean estimated interaction coefficient is stable across dropout cells within each design, indicating no systematic estimand drift.

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- $\hat{\beta}_{bm:D_{bc}}$ shows no systematic trend across none, `MCAR_25`, `biased_25`, and `biased_40` cells within any design.
- Stability holds for the three powered designs (OL+BDC, crossover, hybrid); the OL design is uninformative on bias.
- The absence of estimand drift implies the power degradation traces to standard-error inflation, not bias.

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We turn next to the bias of the biomarker-treatment interaction estimate. The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells within each design, indicating no systematic estimand drift under informative dropout in this regime:

Design	none	MCAR 25%	biased 25%	biased 40%
OL	\$-\$0.09	\$-\$0.10	\$-\$0.09	\$-\$0.10
OL+BDC	\$-\$2.48	\$-\$2.48	\$-\$2.47	\$-\$2.45
traditional crossover	\$-\$2.24	\$-\$2.28	\$-\$2.25	\$-\$2.21
hybrid	\$-\$2.46	\$-\$2.46	\$-\$2.46	\$-\$2.45

[35]

Power degradation under informative dropout traces to standard-error inflation, not estimand drift: empirical SEs expand modestly under biased_40 dropout.

- The largest within-design bias shift (approximately 0.04 for traditional crossover) is within cell-level MCSE.
- Empirical SEs expand under `biased_40`: OL+BDC 0.628 to 0.660, crossover 0.766 to 0.779, hybrid 0.411 to 0.454.
- The power loss in Section 5.1 is therefore attributable to SE inflation rather than systematic bias.

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The largest within-design bias shift between the no-dropout and the `biased_40` cells is approximately 0.04 (traditional crossover), well within the cell-level MCSE on the mean (≤ 0.04 for all designs). The power degradation visible in Section 5.1 therefore traces to inflation of the standard error of $\hat{\beta}_{bm:D_{bc}}$, not to systematic estimand drift. Empirical SEs (last column of `09-dropout-summary.txt`) expand modestly under `biased_40` dropout: from 0.628 to 0.660 for OL+BDC; from 0.766 to 0.779 for traditional crossover; and from 0.411 to 0.454 for hybrid.

6.3 Randomization-path decomposition

[36]

Path-stratified decomposition of the happy-accident effect is scoped for a production rerun; the current prototype provides only marginal results consistent with the hypothesis.

- The current prototype aggregates across paths within each design and cannot attribute robustness to specific surviving paths.
- The Hendrickson happy-accident hypothesis is consistent with the marginal results but requires path-tracked data for a formal test.
- The follow-up extension requires recording `path_id` per replicate and adding path-stratified $\hat{\beta}_{bm:D_{bc}}$ summary logic.

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902 We note that decomposition of the headline power loss into
903 randomisation-path-specific contributions is scoped for a production-
904 grade rerun with per-path-id tracking added to the driver. The
905 current prototype aggregates across paths within each design and
906 therefore cannot directly attribute the hybrid design's robustness
907 to specific paths surviving informative dropout. The substantive¹
908 'happy accident' hypothesis (that informative dropout preferentially
909 preserves the most-informative crossover blocks) is consistent with
910 the present marginal results, but a formal test against the alternative
911 (dropout uniform across paths) requires the path-tracked data that
912 the current prototype does not produce. The follow-up driver exten-
913 sion is relatively straightforward at the simulation level, requiring
914 only that `path_id` be recorded per replicate; the bottleneck is in the
915 analysis layer, where path-stratified $\hat{\beta}_{bm:D_{bc}}$ distributions need to be
916 summarised against the cell-pooled estimate.

917 6.4 Sensitivity to dropout-pattern misspecification

918 [37]

919 **The `biased_25` versus `MCAR_25` contrast isolates the bias mech-**
920 **anism from the rate effect; within-design differences are small**
921 **and statistically indistinguishable at 500 replicates per cell.**

- 922 • OL+BDC: `biased_25` minus `MCAR_25` equals +1.2 pp (CI
923 $[-2.4, +4.8]$); crossover: +1.4 pp (CI $[-3.5, +6.3]$).
- 924 • Hybrid is saturated; OL is at null; all differences lie within or
925 near 1.96 cell-MCSE.
- 926 • Resolving the distinguishability question would require approxi-
927 mately 2,000 replicates per cell at the current MCSE.

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930 The `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout
931 isolates the bias mechanism from the rate effect. The within-design
932 differences are small in absolute terms: OL+BDC `biased_25` power
933 minus `MCAR_25` power equals +1.2 percentage points (CI $[-2.4, +4.8]$);
934 traditional crossover equals +1.4 percentage points (CI $[-3.5, +6.3]$);
935 hybrid is approximately zero (saturated; bracketed); and OL equals
936 -1.2 percentage points (null cells; bracketed).

937 [38]

938 **The dominant power determinant is dropout rate rather than**
939 **bias mechanism; mechanism effects are not statistically dis-**
940 **tinguishable at 500 replicates per cell.**

- Moving from 25% to 40% biased dropout costs 3.0 pp (OL+BDC), 5.6 pp (crossover), 0.6 pp (hybrid).
- The crossover rate-effect loss is clearly detectable above MCSE; the OL+BDC loss is borderline.
- Resolving the `biased_25` versus `MCAR_25` mechanism question requires approximately 2,000 replicates per cell, deferred to a production rerun.

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All four within-design differences lie within 1.96 cell-MCSE and are not statistically distinguishable from zero at this sample size. The dominant effect on power is the dropout rate rather than the bias mechanism: moving from 25 percent to 40 percent biased dropout costs 3.0 percentage points (OL+BDC), 5.6 percentage points (traditional crossover), and 0.6 percentage points (hybrid), with the crossover loss clearly detectable above MCSE. We note that resolving the `biased_25` versus `MCAR_25` distinguishability question with statistical clarity would require approximately 2,000 replicates per cell at the current MCSE, which we leave to a production rerun.

7 Design-by-dropout recommendation

[39]

Three classes of anticipated dropout mechanism are mapped to preferred N-of-1 design choices based on the Section 5 simulation results.

- The mapping is indexed by anticipated overall dropout rate within each mechanism class.
- The recommendation draws directly on the power and bias results from the 16-cell prototype simulation.
- The table is intended as practical guidance for trial designers selecting among the four canonical N-of-1 families.

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In this section we shall map three classes of anticipated dropout mechanism (no informative dropout, informative-by-rate, and informative-by-mechanism) to preferred N-of-1 design choices, drawing on the Section 5 results.

Anticipated dropout	Preferred design	Rationale
None or low MCAR ($d \leq 10\%$)	hybrid	Saturated power; no penalty for hybrid's complexity
Moderate MCAR ($d \approx 25\%$)	hybrid	Hybrid retains saturated power; OL+BDC loses 4 pp; crossover loses 4 pp
Heavy MCAR or biased ($d \geq 40\%$)	hybrid	Hybrid at 99% power; OL+BDC drops to 89%; crossover drops to 77%
Worsening-driven (any rate)	hybrid or OL+BDC	OL+BDC's blinded discontinuation phase carries the bias signal explicitly
Improvement-driven (any rate)	OL+BDC	Crossover loses signal when responders drop early; OL+BDC absorbs
Information-poor (OL only)	not recommended	Architecture-A OL has no within-subject $bm:D_{bc}$ contrast

[40]

The dominant recommendation message is that the hybrid design's power advantage extends to dropout robustness across all three mechanism classes.

- Hybrid is preferred under all dropout rate and mechanism classes except improvement-driven, where OL+BDC is preferred.
- Traditional crossover loses approximately 9 percentage points under `biased_40` and is the least dropout-robust powered design.
- The OL design lacks the within-subject D_{bc} contrast under Architecture A and does not warrant inclusion in the primary recommendation.

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The dominant message from the table is that the hybrid Hendrickson design's power advantage under no dropout extends to robustness against informative dropout. The traditional crossover, despite its strong no-dropout power, loses approximately 9 percentage points under `biased_40` dropout and is the least dropout-robust of the three powered designs. The OL design lacks the within-subject D_{bc} contrast under Architecture A and serves only as a null check at the present prototype's resolution; it therefore does not warrant inclusion in the primary recommendation.

8 Discussion

8.1 Prototype Monte Carlo confirmation

[41]

The 500-replicate prototype confirms 100 percent convergence across all 8,000 fits with achieved dropout fractions close to targets.

- Executed at $N = 35$, $c_{bm} = 0.45$, Architecture A on 8 cores in 421 s with 100 percent convergence.
- Achieved dropout: approximately 0.25 for `MCAR_25` and `biased_25`; approximately 0.40 for `biased_40`.
- Cell MCSE on power runs 0.014–0.019; hybrid saturates near 1.000.

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A 500-replicate-per-cell prototype run on the full 16-cell design-by-dropout grid (design $\in \{\text{OL}, \text{OL+BDC}, \text{traditional crossover}, \text{hybrid}\} \times \text{dropout pattern} \in \{\text{none}, \text{MCAR 25\%}, \text{biased 25\%}, \text{biased 40\%}\}$) was executed at $N = 35$, $c_{bm} = 0.45$ under Architecture A using 8-core `parallel::mclapply`. Wall clock was 421 s; convergence was 100 percent across all 8,000 fits. Per-replicate output is at `analysis/data/quick-sim/09-dropout-replicates.rds`; the Morris ADEMP summary is at `09-dropout-summary.txt`. Achieved dropout fractions tracked targets closely: approximately 0.25 for both `MCAR_25` and `biased_25`, approximately 0.40 for `biased_40`. Cell MCSE on power runs 0.014–0.019 in the binding 0.85 regime, with the hybrid cells effectively saturated near 1.000.

[42]

The prototype reproduces the Hendrickson happy-accident power ordering: hybrid > OL+BDC > crossover > OL at nominal alpha.

- Power ordering: hybrid approximately 1.00, OL+BDC approximately 0.95, crossover approximately 0.85, OL at nominal 0.02–0.05.
- Hybrid dominates because post-week-8 branching produces within-subject and across-path D_{bc} contrasts that survive severe attrition.
- The ordering holds across all four dropout-pattern cells, confirming the Hendrickson happy-accident argument at wider parameter ranges.

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The four designs reproduce the¹ happy-accident randomisation-path ordering at the wider grid: hybrid (approximately 1.00 power) is followed by OL+BDC (approximately 0.95), then traditional crossover (approximately 0.85), then OL (power at the nominal type I level

near 0.02–0.05). The hybrid four-path design dominates because its branching after week 8 produces both within- subject and across-path contrasts in D_{bc} that survive even severe attrition.

[43]

The OL design’s nominal-alpha rejection rate is a structural property of Architecture A, not an artefact of dropout, and serves as an implicit null check.

- Under Architecture A, $D_b \equiv 1$ for all OL timepoints, eliminating any within-subject $bm:D_{bc}$ contrast.
- The fitter has no interaction information, so rejection rates collapse to nominal alpha across all dropout conditions.
- This confirms the OL row is a structural null check rather than evidence of dropout robustness.

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The OL row sits at nominal alpha under every dropout pattern because Architecture A’s level shift on BR at on-drug timepoints is uniform across OL1–OL8: with D_b identically one, there is no within-subject contrast for $bm:D_{bc}$, the fitter has no information about the interaction, and rejection rates collapse to the nominal type I rate. This is a structural property of the OL design under Architecture A, not a dropout artefact; the OL row therefore functions as an implicit null check, not a power estimate.

[44]

Dropout resilience is dominated by the rate effect, not the bias mechanism; the linear-mixed model with corCAR1 is relatively unbiased under informative dropout in this regime.

- Absolute power loss from no-dropout to **biased_40**: hybrid 0.004, OL+BDC 0.056, crossover 0.086.
- The **biased_25** vs. **MCAR_25** mechanism contrast is within or barely outside 1.96 MCSE; the rate increase from 25 to 40 percent is the dominant effect.
- $\hat{\beta}_{bm:D_{bc}}$ is stable (−2.45 to −2.48), confirming power degradation traces to SE inflation rather than estimand drift.

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Resilience to informative dropout, ranked by absolute power loss from the no-dropout cell to the **biased_40** cell, runs as follows: hybrid (loss approximately 0.004) above OL+BDC (loss approximately 0.056) above traditional crossover (loss approximately 0.086). The **biased_25** versus **MCAR_25** contrast at fixed 25 percent dropout is

small (+0.012, +0.014, −0.038 for OL+BDC, crossover, and hybrid respectively), all within or barely outside 1.96 MCSE. The dominant effect is dropout rate, not bias mechanism: moving from 25 percent to 40 percent biased dropout costs 0.006–0.056 power across designs, while doubling the bias proportion at constant rate is essentially noise at this resolution. The mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells (−2.45 to −2.48 for OL+BDC and hybrid), suggesting that the linear-mixed model with `corCAR1` is relatively unbiased under informative dropout in this regime; the power degradation traces to standard-error inflation rather than systematic estimand drift.

[45]

Two ADEMP caveats limit the prototype’s scope: no formal null cells for non-OL designs, and single-architecture coverage.

- No $c_{bm} = 0$ cells were run for OL+BDC, crossover, or hybrid; production reruns should add explicit null cells for each.
- The prototype is single-architecture (Architecture A); under Architecture B the OL row carries information via the covariance structure.
- Resolving the `biased_25` vs. `MCAR_25` distinguishability question requires approximately 2,000 replicates per cell.

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Two ADEMP-discipline caveats warrant mention. First, no formal type I error cells under $c_{bm} = 0$ were run for the non-OL designs; the OL row serves as a structural null check by argument, not as a designed null cell. A canonical production run should add explicit $c_{bm} = 0$ cells for OL+BDC, crossover, and hybrid. Second, the prototype is single-architecture (Architecture A) by specification; under Architecture B (MVN) the OL row would carry information because the biomarker-by-treatment signal lives in the covariance structure rather than in a within-subject mean contrast. Resolving the `biased_25` versus `MCAR_25` distinguishability question would, as noted above, require approximately 2,000 replicates per cell at the current MCSE.

9 Conclusions

[46]

The conclusions distil three findings: design-by-dropout differentiation, rate dominance over mechanism, and SE inflation as the power-loss mechanism.

- The four N-of-1 design families differ materially in dropout robustness in ways the published literature has not characterised.
- Dropout rate dominates mechanism as the determinant of power loss; mechanism becomes load-bearing only when rates are comparable.
- Estimand stability under informative dropout is reassuring but also a warning: coverage degrades as SEs widen without the model acknowledging it.

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In conclusion, three points merit emphasis.

[47]

First conclusion: the hybrid design is the most dropout-robust of the four, losing less than half a percentage point under 40 percent biased dropout at the reference parameters.

- At $N = 35$, $c_{bm} = 0.45$, Architecture A: hybrid loses < 0.5 pp, OL+BDC loses 5.6 pp, crossover loses 8.6 pp under `biased_40`.
- OL lacks the within-subject D_{bc} contrast under Architecture A and is not viable for the biomarker-interaction question.
- The hybrid's robustness extends the Hendrickson happy-accident argument without requiring path-stratified analysis.

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First, the four candidate aggregated N-of-1 trial designs differ in their robustness to informative dropout in ways that the published methodological literature has not adequately characterised. At the prazosin/PTSD reference parameters with $N = 35$, $c_{bm} = 0.45$, and Architecture A, the hybrid Hendrickson design is the most dropout-robust of the four, losing less than half a percentage point of power under 40 percent biased dropout. The OL+BDC design loses approximately 5.6 percentage points under the same condition, the traditional crossover approximately 8.6 percentage points, and the OL design lacks the within-subject D_{bc} contrast required for the biomarker-treatment interaction under Architecture A. The hybrid's robustness extends and corroborates the¹ happy-accident randomisation-path argument without requiring path-stratified analysis to support the substantive claim.

[48]

Second conclusion: dropout rate is the dominant power determinant; mechanism is load-bearing only when rates are comparable across competing design candidates.

- The `biased_25` vs. `MCAR_25` contrast at fixed 25 percent dropout is below MCSE detection at 500 replicates.
- Moving to 40 percent biased dropout produces clear losses in OL+BDC and crossover; hybrid is unaffected.
- Trial designers should evaluate designs against anticipated dropout rate first, and mechanism only secondarily.

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Second, the dominant determinant of the design-by-dropout interaction, in this prototype's evaluation, is the dropout rate rather than the dropout mechanism. The `biased_25` versus `MCAR_25` contrast at fixed 25 percent dropout produces within-design differences below MCSE detection at 500 replicates per cell; moving to 40 percent biased dropout, by contrast, produces clear power losses in every design except the hybrid. Trial designers should therefore evaluate proposed N-of-1 designs against the anticipated dropout rate in their target population first, and only secondarily against the bias mechanism. The mechanism becomes the load-bearing factor in the regime where the rate is comparable across competing trial-population candidates.

[49]

Third conclusion: estimand stability is reassuring but also a coverage warning, since the $\hat{\beta}_{bm:D_{bc}}$ distribution widens without the model acknowledging it.

- The largest within-design shift in $\hat{\beta}_{bm:D_{bc}}$ is below 0.04 across the no-dropout to `biased_40` range.
- Power degradation traces to SE inflation; the inferential target is preserved under informative dropout in this regime.
- Nominal CI coverage may understate true uncertainty: the coefficient distribution widens but the model does not acknowledge it.

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Third, the mean estimated coefficient $\hat{\beta}_{bm:D_{bc}}$ is stable across dropout cells, with the largest within-design shift below 0.04 across the no-dropout to `biased_40` range. The power degradation under informative dropout therefore traces to standard-error inflation rather than systematic estimand drift. This finding is reassuring for the validity of the aggregated N-of-1 framework in the presence of informative dropout, in the sense that the inferential target is preserved, but it is also a warning to trialists that nominal coverage of confidence intervals may understate the true uncertainty in the dropout regime, since the $\hat{\beta}_{bm:D_{bc}}$ distribution widens without the model acknowledging it.

Three production-grade extensions are scoped: path tracking, Architecture B coverage, and higher replicate counts to resolve the biased-versus-MCAR distinguishability question.

- Per-path-id tracking is needed for the randomisation-path decomposition of the happy-accident effect.
- Architecture B coverage is needed to interpret the OL row as a power estimate rather than a structural null check.
- Resolving the biased_25 vs. MCAR_25 question requires 2,000 or more replicates per cell.

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Production-grade extensions of the prototype (per-path tracking for the randomisation-path decomposition, Architecture B for the OL design's null-check elevation, and 2,000 or more replicates per cell to resolve the biased-versus-MCAR distinguishability question) are scoped in Section 5 of this manuscript and represent the natural follow-up to the present proof-of-concept evidence.

10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's `zzcollab` Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the original-extended simulation collection), `furrr` and `purrr` (the `tidyverse`

simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.
 The informative-dropout module is at `R/censordata.R`.
 Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop
 and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive
 from the master seed by `+ rep_idx`.
 Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/09-dropout-replicates.rds`
 Morris ADEMP cell-level summaries are at the companion `09-dropout-summary.txt`.
 Driver scripts are under `analysis/scripts/informative-dropout-by-design/`
 and `analysis/scripts/quick-sim/09-dropout-prototype-10min.R`. The pre-
 registered ADEMP plan is at `analysis/scripts/informative-dropout-by-design/00-ademp-pre-registra`
 The manuscript source, paper-specific bibliography (`references.bib`), and SIM
 CSL file are at `analysis/report/09-informative-dropout-by-design/`.

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